5.2 Annihilators

Section 1 built the dual space V^*, the space of linear measurements on V. Now we connect subspaces of V with subspaces of V^*. A plane in $\mathbb{R}^3$ can be described by a spanning list or by an equation, and an equation is a functional. This section collects all the equations that a set of vectors satisfies into one subspace of V^*, the annihilator, counts its dimension, and shows that it turns sums into intersections and intersections into sums.

5.2.1 The equations a subspace satisfies

Every object comes with its subobjects. The dual space V^* has subspaces of its own, and the question is which of them are attached to subspaces of V. The standard example shows the way.

Let $U = \operatorname{span}((1,0,1),(0,1,1))$, a plane in $\mathbb{R}^3$. The functional $\varphi(x,y,z) = x+y-z$ vanishes on both spanning vectors, so by linearity it vanishes on all of U: the plane satisfies the equation $x+y-z=0$. So do $2x+2y-2z=0$ and the trivial equation $0=0$. Which functionals vanish on U? By Theorem 5.1.1 a functional is $\psi(x,y,z) = ax+by+cz$. It vanishes on the two spanning vectors if and only if $a+c=0$ and $b+c=0$, that is, $\psi = -c(x+y-z) = -c\,\varphi$. So the equations of the plane form the **line** $\operatorname{span}(\varphi)$ in $(\mathbb{R}^3)^*$.

For the z-axis $\operatorname{span}((0,0,1))$, the same computation gives $c=0$: the equations are $ax+by=0$, a **plane** in $(\mathbb{R}^3)^*$. A bigger subspace satisfies fewer equations, and in both cases the dimensions add up to 3. The set of equations deserves a name, and the pattern deserves a theorem.

The annihilator of a set of vectors is the space of all linear equations that every one of those vectors satisfies.

Definition 5.2.1. Annihilator

Let V be a vector space over F, and let $S \subseteq V$ be **any subset**. The **annihilator** of S is

$$S^0 := \{\varphi \in V^* : \varphi(s) = 0 \text{ for all } s \in S\}.$$

In words: S^0 is a subset of the **dual space** V^*, not of V. A functional belongs to it when it vanishes at **every** vector of S; a single nonzero value disqualifies it. The set S needs no structure: it may be a subspace, a finite list of vectors, or even empty. The superscript 0 recalls the value that the functionals take on S.

Well-definedness. The definition makes sense for every subset. What needs checking is that the result is a subspace, which is what makes its dimension meaningful.

Proposition 5.2.1. The Annihilator Is a Subspace

Let V be a vector space over F, and let $S \subseteq V$. Then S^0 is a subspace of V^*.

Proof. We use the subspace test (Theorem 1.2.1) in V^*. The zero functional sends every $s \in S$ to 0, so it lies in S^0. Let $\varphi, \psi \in S^0$ and $c \in F$. For every $s \in S$, the pointwise operations of Definition 5.1.2 give

$$(\varphi+\psi)(s) = \varphi(s)+\psi(s) = 0+0 = 0, \qquad (c\varphi)(s) = c\,\varphi(s) = c\cdot 0 = 0.$$

Hence $\varphi + \psi \in S^0$ and $c\varphi \in S^0$, and S^0 is a subspace of V^*.

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Example. First Annihilators

(a) For any vector space V, find $\{0\}^0$, $\varnothing^0$, and V^0.

(b) In $M_2(F)$, find the annihilator of the single matrix $\mathbf{E}_{11}$.

(c) In $\mathbb{R}[x]_{\leq 2}$, find the annihilator of $S = \{1, x\}$.

Solution.

(a) Every functional sends 0 to 0 (Theorem 2.1.1), so $\{0\}^0 = V^*$. The condition "for all $s \in \varnothing$" holds vacuously, so $\varnothing^0 = V^*$. A functional in V^0 takes the value 0 at every vector, so it is the zero functional, and $V^0 = \{0\}$. These degenerate cases fix the direction of everything that follows: the smallest subspace of V has the largest annihilator, and the largest has the smallest.

(b) Let $\varepsilon_{ij}(\mathbf{A}) = a_{ij}$ be the dual basis of the standard basis $(\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$. By Theorem 5.1.3 (c), every $\psi \in M_2(F)^*$ is $\psi = \sum_{i,j} \psi(\mathbf{E}_{ij})\, \varepsilon_{ij}$. Such a ψ lies in $\{\mathbf{E}_{11}\}^0$ exactly when $\psi(\mathbf{E}_{11}) = 0$. Hence

$$\{\mathbf{E}_{11}\}^0 = \operatorname{span}(\varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22}),$$

the functionals $\mathbf{A} \mapsto c_{12}a_{12} + c_{21}a_{21} + c_{22}a_{22}$. It has dimension 3, and $3 + 1 = 4 = \dim M_2(F)$.

(c) Let $(\varphi_0, \varphi_1, \varphi_2)$ be the dual basis of $(1, x, x^2)$, so φ_k reads off the coefficient of x^k. A functional $\psi = \psi(1)\varphi_0 + \psi(x)\varphi_1 + \psi(x^2)\varphi_2$ lies in S^0 exactly when $\psi(1) = \psi(x) = 0$, that is, when $\psi = \psi(x^2)\varphi_2$. So $S^0 = \operatorname{span}(\varphi_2)$: the only equations satisfied by both 1 and x are the multiples of "the coefficient of x^2 is 0".

Nonexample by minimal change. Shift the plane of the opening example off the origin: $A = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 1\}$. The set is still cut out by the same left-hand side, but $\varphi(x, y, z) = x + y - z$ takes the value 1, not 0, on A, so the defining clause $\varphi(s) = 0$ fails and $\varphi \notin A^0$. In fact $A^0 = \{0\}$: the set A contains $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, -1)$, and a functional $ax + by + cz$ vanishing at these three vectors has $a = b = c = 0$. Annihilators record only **homogeneous** equations.

Why this definition. We allow any subset S, not only subspaces, because subspaces are usually handed to us by spanning lists, and we want to test a functional on the list alone. The theorem at the end of this section shows that this loses nothing: $S^0 = (\operatorname{span} S)^0$. The name is literal. To annihilate is to reduce to nothing, and the functionals in S^0 reduce every vector of S to 0.

Warning. U^0 **lives in** V^***, and it is not a complement of** U**.** It is tempting to turn the functional $ax+by$ into the vector (a, b) and to picture U^0 as a subspace of F^2 "perpendicular" to U. Over $\mathbb{F}_2$ this picture fails completely. Let $U = \operatorname{span}((1, 1)) \subseteq \mathbb{F}_2^2$. A functional $ax + by$ vanishes on $(1, 1)$ exactly when $a + b = 0$, that is, $a = b$, so $U^0 = \{0,\ x + y\}$. The functional $x+y$ corresponds to the vector $(1, 1)$, which lies **in** U. So the "picture" of U^0 is U itself, not a

complement. Perpendicularity needs an inner product, which is the orthogonal complement of Chapter 11; the annihilator needs nothing.

The payoff of the definition is the pattern we saw in the opening example. The equations of a subspace take up exactly the dimensions that the subspace leaves free.

5.2.2 Counting equations

Theorem 5.2.2. Dimension of the Annihilator

Let V be a finite-dimensional vector space over F, and let U be a subspace of V. Then

$$\dim U + \dim U^0 = \dim V.$$

More precisely, if $(u_1, \dots, u_m, v_{m+1}, \dots, v_n)$ is a basis of V extending a basis $(u_1, \dots, u_m)$ of U, with dual basis $(\varphi_1, \dots, \varphi_n)$, then $(\varphi_{m+1}, \dots, \varphi_n)$ is a basis of U^0.

Idea. This is a dimension formula, so we start from the smallest subspace, U, take a basis and extend it to V. The dual basis now splits in two. The last $n - m$ functionals vanish on every u_i, by the delta, so they vanish on U. The first m do not. To see that nothing else is in U^0, expand a functional $\psi \in U^0$ in the dual basis: its coefficients are its values on the basis, and the first m values are $\psi(u_i) = 0$.

Proof. Since U is a subspace of the finite-dimensional space V, it has a basis $(u_1, \dots, u_m)$ (Theorem 1.6.7, Corollary 1.5.5), and by the Basis Extension Theorem (Theorem 1.6.5) it extends to a basis $\mathcal{B} = (u_1, \dots, u_m, v_{m+1}, \dots, v_n)$ of V, where $n = \dim V$. Let $(\varphi_1, \dots, \varphi_n)$ be the dual basis of $\mathcal{B}$ (Theorem 5.1.3).

The φ_j with $j > m$ lie in U^0. Let $j > m$ and $u \in U$, say $u = a_1 u_1 + \cdots + a_m u_m$. By Theorem 2.1.4, $\varphi_j(u) = \sum_{i=1}^{m} a_i \varphi_j(u_i) = 0$, since $\varphi_j(u_i) = \delta_{ji} = 0$ for $i \le m < j$.

They span U^0. Let $\psi \in U^0$. By Theorem 5.1.3 (c),

$$\psi = \sum_{i=1}^{m} \psi(u_i)\, \varphi_i + \sum_{j=m+1}^{n} \psi(v_j)\, \varphi_j = \sum_{j=m+1}^{n} \psi(v_j)\, \varphi_j,$$

because $\psi(u_i) = 0$ for $u_i \in U$. Hence $U^0 \subseteq \operatorname{span}(\varphi_{m+1}, \dots, \varphi_n)$, and the reverse inclusion holds because U^0 is a subspace (Proposition 5.2.1, Theorem 1.3.1).

They are independent, being part of the basis $(\varphi_1, \dots, \varphi_n)$ of V^*. Therefore $(\varphi_{m+1}, \dots, \varphi_n)$ is a basis of U^0, and $\dim U^0 = n - m = \dim V - \dim U$. This proves the theorem.

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Alternatively, restrict functionals to U. The map $R\colon V^* \to U^*$, $\varphi \mapsto \varphi|_U$, is linear, and its kernel is U^0 by definition. It is surjective: given $\xi \in U^*$, Theorem 2.1.5 gives $\varphi \in V^*$ with $\varphi(u_i) = \xi(u_i)$ and $\varphi(v_j) = 0$, and then $\varphi|_U = \xi$ because both agree on a basis of U. By the Rank–Nullity Theorem (Theorem 2.3.1) and Corollary 5.1.4, $\dim V = \dim V^* = \dim U^0 + \dim U^* =$

$\dim U^0 + \dim U$. This route shows **why** the formula holds: a functional on V is a functional on U plus a choice of values off U, and U^0 measures that choice.

The theorem is the second signature move of the chapter: **pass to the dual and count.** One immediate consequence: if U is a **proper** subspace of a finite-dimensional V, then $\dim U^0 > 0$, so some **nonzero** functional vanishes on U. Equivalently, a list spans a finite-dimensional V if and only if the only functional that kills every vector of the list is 0: apply the previous sentence to the span U of the list, and note that a functional kills every vector of the list exactly when it kills U, by Theorem 2.1.4.

In F^n the whole computation becomes elimination, because functionals are rows.

Example. An Annihilator as a Null Space

Find a basis of U^0 for $U = \operatorname{span}((1,2,0),(0,1,1)) \subseteq \mathbb{R}^3$, and check the dimension formula.

Solution. By Theorem 5.1.1, every functional is $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ for a unique $\mathbf{a} = (a_1, a_2, a_3)$. By linearity, $\varphi_{\mathbf{a}}$ vanishes on U if and only if it vanishes on the two spanning vectors (a combination of vectors killed by $\varphi_{\mathbf{a}}$ is killed, by Theorem 2.1.4). These two conditions are

$$a_1 + 2a_2 = 0, \qquad a_2 + a_3 = 0,$$

$$\text{that is,} \qquad \mathbf{A}\mathbf{a} = \mathbf{0} \quad \text{with} \quad \mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

the matrix whose **rows** are the spanning vectors. So $\varphi_{\mathbf{a}} \in U^0$ if and only if $\mathbf{a} \in \operatorname{nul}(\mathbf{A})$. Subtracting twice row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix},$$

with a_3 free. By Theorem 3.5.4, $\operatorname{nul}(\mathbf{A}) = \operatorname{span}((2,-1,1))$. Hence $U^0 = \operatorname{span}(\psi)$ with

$$\psi(x,y,z) = 2x - y + z.$$

Check: $\psi(1,2,0) = 2 - 2 + 0 = 0$ and $\psi(0,1,1) = -1 + 1 = 0$. The two spanning vectors are independent (neither is a multiple of the other), so $\dim U = 2$, and $\dim U + \dim U^0 = 2 + 1 = 3$, as Theorem 5.2.2 predicts. As a by-product, U is the plane $2x - y + z = 0$: $U \subseteq \ker \psi$, both have dimension 2 (Proposition 5.1.2), and Theorem 1.6.8 gives equality.

The same recipe works in general. For $U = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_m) \subseteq F^n$, let $\mathbf{A}$ have rows $\mathbf{u}_1^\top, \ldots, \mathbf{u}_m^\top$. Then $U^0 = \{\varphi_{\mathbf{a}} : \mathbf{a} \in \operatorname{nul}(\mathbf{A})\}$, and the isomorphism of Theorem 5.1.1 carries a basis of $\operatorname{nul}(\mathbf{A})$ to a basis of U^0.

Away from F^n, counting often replaces the computation entirely.

Example. The Equations of a Root Condition

Let $U = \{p \in \mathbb{R}[x]_{\le 2} : p(1) = 0\}$. Find U^0.

Solution. Let $\varepsilon_1(p) = p(1)$, a linear functional (Example (Four Families of Functionals)). By definition $U = \ker \varepsilon_1$, so $\varepsilon_1 \in U^0$. Since $\varepsilon_1(1) = 1$, $\varepsilon_1 \neq 0$, and Proposition 5.1.2 gives $\dim U = 3 - 1 = 2$. By Theorem 5.2.2, $\dim U^0 = 3 - 2 = 1$. Now $\operatorname{span}(\varepsilon_1)$ is a subspace of U^0 (Proposition 5.2.1) of dimension 1, so $U^0 = \operatorname{span}(\varepsilon_1)$ by Theorem 1.6.8.

We never found a basis of U and never solved an equation. The conclusion is concrete: a functional on $\mathbb{R}[x]_{\leq 2}$ vanishes on every polynomial with a root at 1 **only if** it is a multiple of $p \mapsto p(1)$.

Quick check. Let $U = \operatorname{span}((1,1,1)) \subseteq \mathbb{R}^3$. What is $\dim U^0$? Name a basis of U^0 without solving a system.

Solution. $\dim U^0 = 3 - 1 = 2$ by Theorem 5.2.2. The functionals $x - y$ and $y - z$ vanish at $(1,1,1)$, hence on U, and they are independent: in $a(x-y) + b(y-z) = 0$, the coefficient of x gives $a = 0$, and then $b = 0$. Two independent vectors in a 2-dimensional space form a basis (Theorem 1.6.4).

5.2.3 Sums, intersections, and spans

Taking annihilators reverses inclusions: more vectors satisfy fewer common equations. It also exchanges sums with intersections. In $\mathbb{R}^3$, let U be the xy-plane and W the xz-plane. Then $U^0 = \operatorname{span}(z)$ and $W^0 = \operatorname{span}(y)$, writing x, y, z for the coordinate functionals. The intersection $U \cap W$ is the x-axis, with annihilator $\operatorname{span}(y, z) = U^0 + W^0$. The sum $U + W = \mathbb{R}^3$ has annihilator $\{0\} = U^0 \cap W^0$. The theorem says this always happens.

Theorem 5.2.3. Properties of Annihilators

Let V be a vector space over F.

(a) If $S \subseteq T \subseteq V$, then $T^0 \subseteq S^0$.

(b) For every subset $S \subseteq V$, $S^0 = (\operatorname{span} S)^0$.

(c) For subspaces U, W of V, $(U + W)^0 = U^0 \cap W^0$.

(d) If V is **finite-dimensional**, then for subspaces U, W of V, $(U \cap W)^0 = U^0 + W^0$.

Idea. Parts (a)–(c) are unwinding definitions: fix a functional in one side and test it on the vectors of the other. In (d), the inclusion $U^0 + W^0 \subseteq (U \cap W)^0$ is also direct. The reverse inclusion is the hard half. Given φ vanishing on $U \cap W$, we would have to **split** it as $\psi + \chi$ with ψ killing U and χ killing W, and nothing in sight tells us how. So we count instead: one inclusion plus equal dimensions gives equality. The plan is ① compute $\dim(U^0 + W^0)$ by the dimension formula for sums inside V^*; ② replace $U^0 \cap W^0$ by $(U + W)^0$ using (c); ③ turn every annihilator dimension into a dimension in V with Theorem 5.2.2; ④ recognize the dimension formula in V. Finite dimension is spent in step ③.

Proof.

(a) Let $\varphi \in T^0$. Every $s \in S$ lies in T, so $\varphi(s) = 0$. Hence $\varphi \in S^0$.

(b) Since $S \subseteq \operatorname{span} S$ (Theorem 1.3.1), part (a) gives $(\operatorname{span} S)^0 \subseteq S^0$. Conversely, let $\varphi \in S^0$ and $v \in \operatorname{span} S$, say $v = a_1 s_1 + \cdots + a_k s_k$ with $s_i \in S$ and $k \geq 0$. By Theorem 2.1.4, $\varphi(v) = \sum_i a_i \varphi(s_i) = 0$. Hence $\varphi \in (\operatorname{span} S)^0$.

(c) Since $U \subseteq U + W$ and $W \subseteq U + W$ (Theorem 1.7.1), part (a) gives $(U+W)^0 \subseteq U^0$ and $(U+W)^0 \subseteq W^0$, so $(U+W)^0 \subseteq U^0 \cap W^0$. Conversely, let $\varphi \in U^0 \cap W^0$, and let $u + w \in U + W$ with $u \in U$, $w \in W$. Then $\varphi(u+w) = \varphi(u) + \varphi(w) = 0 + 0 = 0$. Hence $\varphi \in (U+W)^0$.

(d) ($\supseteq$) Let $\varphi = \psi + \chi$ with $\psi \in U^0$ and $\chi \in W^0$, and let $v \in U \cap W$. Since $v \in U$ and $v \in W$, $\varphi(v) = \psi(v) + \chi(v) = 0 + 0 = 0$. Hence $U^0 + W^0 \subseteq (U \cap W)^0$.

($\subseteq$) Let $n = \dim V$. The subspaces U, W, $U \cap W$ (Theorem 1.2.2), and $U + W$ (Theorem 1.7.1) of V are finite-dimensional by Theorem 1.6.7. The space V^* has dimension n by Corollary 5.1.4, so its subspaces U^0, W^0 are finite-dimensional too. By the Dimension Formula for Sums (Theorem 1.7.3) in V^*, then (c),

$$\begin{aligned} \dim(U^0 + W^0) &= \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) \\ &= \dim U^0 + \dim W^0 - \dim(U+W)^0. \end{aligned}$$

By Theorem 5.2.2 applied to U, W, and $U + W$, the right-hand side is

$$\begin{aligned} &(n - \dim U) + (n - \dim W) - (n - \dim(U+W)) \\ &\quad = n - (\dim U + \dim W - \dim(U+W)) \\ &\quad = n - \dim(U \cap W), \end{aligned}$$

where the last equality is Theorem 1.7.3 in V. By Theorem 5.2.2 once more, $n - \dim(U \cap W) = \dim(U \cap W)^0$. So $U^0 + W^0$ is a subspace of $(U \cap W)^0$ (by ($\supseteq$) and Theorem 1.7.1) with the same finite dimension, and Theorem 1.6.8 gives $U^0 + W^0 = (U \cap W)^0$. This proves the theorem.

■

Part (b) is what made the computations of this section legitimate: to find the annihilator of a subspace, test functionals on a spanning list only. Parts (c) and (d) form a dictionary. A subspace given as a **sum** has equations obtained by **intersecting** equation spaces; a subspace given as an **intersection** has equations obtained by **adding** them. Later in this chapter this dictionary becomes a practical method for computing sums and intersections of subspaces.

Remark. Parts (a)–(c) and the inclusion ($\supseteq$) in (d) hold in every vector space; only the counting step used finite dimension. The equality (d) is in fact true for **every** V, but the proof must then construct the splitting $\varphi = \psi + \chi$ by hand; Exercise C1 below does this. In finite dimension, (a) also becomes strict: if $U \subsetneq W$, then $\dim U < \dim W$ (Theorem 1.6.8), so $\dim W^0 < \dim U^0$ and $W^0 \subsetneq U^0$.

5.2.4 Exercises

5.2.4.1 A. Check your understanding

Exercise. A1

(a) Define the annihilator S^0 of a subset S of a vector space V.

(b) True or false: U^0 is a subspace of V. Justify your answer.

(c) Let U be a 2-dimensional subspace of $\mathbb{R}^5$. What is $\dim U^0$?

(d) For a finite-dimensional V, what are $\{0\}^0$ and V^0?

(e) True or false: if $U \subsetneq W$ are subspaces of a finite-dimensional V, then $W^0 \subsetneq U^0$. Justify your answer.

(f) To compute U^0 for $U = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_m) \subseteq F^n$, which matrix do you row reduce, and why?

Solution.

(a) $S^0 = \{\varphi \in V^* : \varphi(s) = 0 \text{ for all } s \in S\}$ (Definition 5.2.1).

(b) False. U^0 is a subspace of the dual space V^* (Proposition 5.2.1); its elements are functionals, not vectors of V.

(c) $\dim U^0 = 5 - 2 = 3$, by Theorem 5.2.2.

(d) $\{0\}^0 = V^*$ and $V^0 = \{0\}$, by Example (First Annihilators) (a).

(e) True. By Theorem 5.2.3 (a), $W^0 \subseteq U^0$. Since $U \subsetneq W$, $\dim U < \dim W$ by Theorem 1.6.8, so $\dim W^0 = \dim V - \dim W < \dim V - \dim U = \dim U^0$. Hence the inclusion is strict.

(f) The matrix $\mathbf{A}$ whose **rows** are $\mathbf{u}_1^\top, \ldots, \mathbf{u}_m^\top$. By Theorem 5.1.1 a functional is $\mathbf{x} \mapsto \mathbf{a}^\top \mathbf{x}$, and it kills every $\mathbf{u}_i$ exactly when $\mathbf{A}\mathbf{a} = \mathbf{0}$; so U^0 corresponds to $\operatorname{nul}(\mathbf{A})$ (Example (An Annihilator as a Null Space)).

5.2.4.2 B. Practice

Exercise. B1: Computing annihilators

(a) Let $U = \operatorname{span}((1,2,1,0), (2,4,3,1), (1,2,2,1)) \subseteq \mathbb{R}^4$. Find a basis of U^0, and verify that $\dim U + \dim U^0 = 4$.

(b) Let $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$. Show that $U^0 = \operatorname{span}(\varepsilon_1, \varepsilon_{-1})$, where $\varepsilon_c(p) = p(c)$.

Solution.

(a) Let $\mathbf{A}$ have the three spanning vectors as rows. As in Example (An Annihilator as a Null Space), $\varphi_{\mathbf{a}} \in U^0$ if and only if $\mathbf{A}\mathbf{a} = \mathbf{0}$. Subtracting 2 times row 1 from row 2 and row 1 from row 3 gives rows $(1,2,1,0)$, $(0,0,1,1)$, $(0,0,1,1)$; subtracting row 2 from

row 3 and then row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are a_2 and a_4, and Theorem 3.5.4 gives

$$\operatorname{nul}(\mathbf{A}) = \operatorname{span}((-2,1,0,0),(1,0,-1,1)).$$

Hence U^0 has the basis

$$\psi_1(x_1,x_2,x_3,x_4) = -2x_1 + x_2, \qquad \psi_2(x_1,x_2,x_3,x_4) = x_1 - x_3 + x_4.$$

Check on the first two spanning vectors: $\psi_1(1,2,1,0) = 0$, $\psi_2(1,2,1,0) = 0$, $\psi_1(2,4,3,1) = 0$, $\psi_2(2,4,3,1) = 2-3+1 = 0$; the third vector is their difference, $(2,4,3,1) - (1,2,1,0)$. The RREF has two pivots, so $\dim U = \dim \operatorname{row}(\mathbf{A}) = 2$ (Theorem 3.5.5), and $2+2=4$.

(b) By definition $\varepsilon_1, \varepsilon_{-1} \in U^0$. *Independent:* if $a\varepsilon_1 + b\varepsilon_{-1} = 0$, applying it to $x+1$ gives $2a = 0$ and applying it to $x-1$ gives $-2b = 0$. *Dimension of U:* the map $T\colon \mathbb{R}[x]_{\le 3} \to \mathbb{R}^2$, $p \mapsto (p(1), p(-1))$, is linear with kernel U, and its image contains $T(x+1) = (2,0)$ and $T(x-1) = (0,-2)$, which span $\mathbb{R}^2$. So $\operatorname{rank} T = 2$ and $\dim U = 4 - 2 = 2$ by Theorem 2.3.1. By Theorem 5.2.2, $\dim U^0 = 2$. Hence the 2-dimensional subspace $\operatorname{span}(\varepsilon_1, \varepsilon_{-1})$ of U^0 is all of U^0 (Theorem 1.6.8).

Exercise. B2: Equations of the symmetric matrices

Let $U = \{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A}^\top = \mathbf{A}\}$. Find U^0, and verify the dimension formula.

Solution. Every symmetric matrix is $a\mathbf{E}_{11} + b(\mathbf{E}_{12} + \mathbf{E}_{21}) + c\mathbf{E}_{22}$, so $U = \operatorname{span}(\mathbf{E}_{11}, \mathbf{E}_{12} + \mathbf{E}_{21}, \mathbf{E}_{22})$. Let (ε_{ij}) be the dual basis of the standard basis, so that every $\psi \in M_2(\mathbb{R})^*$ is $\psi = \sum_{i,j} c_{ij}\varepsilon_{ij}$ with $c_{ij} = \psi(\mathbf{E}_{ij})$ (Theorem 5.1.3 (c)). By Theorem 5.2.3 (b), $\psi \in U^0$ if and only if ψ kills the three spanning matrices:

$$c_{11} = 0, \qquad c_{12} + c_{21} = 0, \qquad c_{22} = 0.$$

So $\psi = c_{12}(\varepsilon_{12} - \varepsilon_{21})$, and $U^0 = \operatorname{span}(\varepsilon_{12} - \varepsilon_{21})$: the only equation of the symmetric matrices, up to scaling, is $a_{12} - a_{21} = 0$. The three spanning matrices are independent (they involve disjoint sets of entries), so $\dim U = 3$, and $\dim U + \dim U^0 = 3 + 1 = 4 = \dim M_2(\mathbb{R})$.

Exercise. B3: The equations of a hyperplane

Let V be a finite-dimensional vector space over F, and let $\varphi \in V^*$ be nonzero. Prove that $(\ker \varphi)^0 = \operatorname{span}(\varphi)$.

Solution. By definition of the kernel, φ vanishes on $\ker \varphi$, so $\varphi \in (\ker \varphi)^0$, and $\operatorname{span}(\varphi) \subseteq (\ker \varphi)^0$ by Proposition 5.2.1. Since $\varphi \neq 0$, $\dim \operatorname{span}(\varphi) = 1$, and Proposition 5.1.2 gives

$\dim \ker \varphi = \dim V - 1$. By Theorem 5.2.2, $\dim(\ker \varphi)^0 = \dim V - (\dim V - 1) = 1$. A 1-dimensional subspace of a 1-dimensional space is the whole space (Theorem 1.6.8), so $(\ker \varphi)^0 = \operatorname{span}(\varphi)$.

5.2.4.3 C. Going deeper

Exercise. C1: The intersection formula without counting

Let U, W be subspaces of a vector space V over F, and let $\varphi \in (U \cap W)^0$.

(a) Suppose V is finite-dimensional. Construct $\psi \in U^0$ and $\chi \in W^0$ with $\varphi = \psi + \chi$, without using any dimension formula.

(b) Explain how to modify the construction when V is infinite-dimensional. Deduce that $(U \cap W)^0 = U^0 + W^0$ holds in every vector space.

Hint: extend a basis of $U \cap W$ to a basis of U and, separately, to a basis of W.

Solution.

(a) Let $(x_1, \dots, x_k)$ be a basis of $U \cap W$ (Theorem 1.2.2, Theorem 1.6.7). Extend it to a basis $(x_1, \dots, x_k, u_1, \dots, u_p)$ of U and to a basis $(x_1, \dots, x_k, w_1, \dots, w_q)$ of W (Theorem 1.6.5).
The combined list $(x_1, \dots, x_k, u_1, \dots, u_p, w_1, \dots, w_q)$ is independent. Let $\sum a_i x_i + \sum b_i u_i + \sum c_j w_j = 0$. Then $\sum c_j w_j = -\sum a_i x_i - \sum b_i u_i$ lies in W and in U, hence in $U \cap W$, so $\sum c_j w_j = \sum d_i x_i$ for some $d_i \in F$. Since (x, w) is a basis of W, all c_j and d_i are 0. Then $\sum a_i x_i + \sum b_i u_i = 0$, and since (x, u) is a basis of U, all a_i and b_i are 0.
Extend the combined list to a basis of V by vectors $y_1, \dots, y_r$ (Theorem 1.6.5). By Theorem 2.1.5 there is $\psi \in V^*$ with

$$\psi(x_i) = 0, \qquad \psi(u_i) = 0, \qquad \psi(w_j) = \varphi(w_j), \qquad \psi(y_l) = 0.$$

Then ψ kills a basis of U, so $\psi \in U^0$ (Theorem 5.2.3 (b)). Put $\chi := \varphi - \psi$. Then $\chi(x_i) = \varphi(x_i) - 0 = 0$, because $x_i \in U \cap W$ and $\varphi \in (U \cap W)^0$; and $\chi(w_j) = \varphi(w_j) - \varphi(w_j) = 0$. So χ kills a basis of W, and $\chi \in W^0$. Hence $\varphi = \psi + \chi \in U^0 + W^0$.

(b) Replace the finite lists by sets. By Theorem 1.8.3, $U \cap W$ has a basis X, and by Theorem 1.8.2 it extends to a basis $X \cup P$ of U and to a basis $X \cup Q$ of W, with $P \cap X = Q \cap X = \varnothing$. First, $P \cap Q = \varnothing$: a vector of $P \cap Q$ lies in $U \cap W = \operatorname{span} X$, which contradicts the independence of $X \cup P$. So every vector of $X \cup P \cup Q$ lies in exactly one of X, P, Q. A linear relation among finitely many distinct vectors of $X \cup P \cup Q$ splits into an X-part, a P-part, and a Q-part, and the independence argument of (a) applies word for word; so $X \cup P \cup Q$ is independent. By Theorem 1.8.2 it extends to a basis B of V. Every $v \in V$ is a finite combination $v = \sum_b c_b b$ of distinct vectors of B, with unique coefficients because B is independent. Define

$$\psi(v) := \sum_{b \in Q} c_b \, \varphi(b),$$

a finite sum over the vectors of Q that occur. Adding or scaling expansions gives expansions, so by uniqueness the coefficients add and scale, and ψ is linear. It vanishes on $X \cup P$ and agrees with φ on Q, so the rest of (a) is unchanged: $\psi \in U^0$ and $\chi = \varphi - \psi$

kills the basis $X \cup Q$ of W. Together with the inclusion ($\supseteq$) of Theorem 5.2.3 (d), which used no finiteness, this gives $(U \cap W)^0 = U^0 + W^0$ in every vector space.

Exercise. C2: Equations of a double root

Let $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = 0 \text{ and } p'(1) = 0\}$, and let $\varepsilon_1(p) = p(1)$ and $\eta_1(p) = p'(1)$. Prove that $U^0 = \operatorname{span}(\varepsilon_1, \eta_1)$. Deduce that $\psi(p) = p(0) - p(2) + 2p'(1)$ does **not** vanish on U.

Solution. Both ε_1 and η_1 are linear (evaluation, and differentiation followed by evaluation), and they vanish on U by definition. *Independent:* if $a\varepsilon_1 + b\eta_1 = 0$, applying it to 1 gives $a = 0$, and applying it to $x-1$ gives $b = 0$. *Dimension of U:* the linear map $T\colon \mathbb{R}[x]_{\leq 3} \to \mathbb{R}^2$, $p \mapsto (p(1), p'(1))$, has kernel U, and $T(1) = (1,0)$, $T(x-1) = (0,1)$, so T is surjective. By Theorem 2.3.1, $\dim U = 4 - 2 = 2$, and by Theorem 5.2.2, $\dim U^0 = 2$. The 2-dimensional subspace $\operatorname{span}(\varepsilon_1, \eta_1)$ of U^0 is therefore all of U^0 (Theorem 1.6.8).

For the deduction, suppose $\psi \in U^0$. Then $\psi = a\varepsilon_1 + b\eta_1$. Applying both sides to 1 gives $1 - 1 + 0 = a$, so $a = 0$; applying them to $x - 1$ gives $-1 - 1 + 2 = b$, so $b = 0$. Then $\psi = 0$. But $\psi(x^2) = 0 - 4 + 4 = 0$ and $\psi(x^3) = 0 - 8 + 6 = -2 \neq 0$, a contradiction. Hence $\psi \notin U^0$. Check with an explicit witness: $p = x(x-1)^2 \in U$ has $\psi(p) = 0 - 2 + 0 = -2 \neq 0$.

Exercise. C3: Dual of a direct sum decomposition

(a) Let V be finite-dimensional with $V = U \oplus W$. Prove that $V^* = U^0 \oplus W^0$.

(b) In $\mathbb{R}^2$, let $U = \operatorname{span}((1,0))$ and $W = \operatorname{span}((1,1))$. Write $\psi(x,y) = x$ as $\psi_U + \psi_W$ with $\psi_U \in U^0$ and $\psi_W \in W^0$.

Solution.

(a) Since the sum is direct, $U \cap W = \{0\}$ (Theorem 1.7.4), and $U + W = V$. By Theorem 5.2.3 (c) and Example (First Annihilators) (a), $U^0 \cap W^0 = (U+W)^0 = V^0 = \{0\}$. By Theorem 5.2.3 (d), $U^0 + W^0 = (U \cap W)^0 = \{0\}^0 = V^*$. So $V^* = U^0 + W^0$, and the sum is direct by Theorem 1.7.4.

(b) A functional $ax + by$ vanishes on $(1,0)$ if and only if $a = 0$, so $U^0 = \operatorname{span}(y)$. It vanishes on $(1,1)$ if and only if $b = -a$, so $W^0 = \operatorname{span}(x - y)$. Solving $x = s\,y + t(x - y)$ gives $t = 1$ and $s - t = 0$, so $s = 1$:

$$\psi = \underbrace{y}_{\in U^0} + \underbrace{(x - y)}_{\in W^0}.$$

By (a) this decomposition is unique.